Wk 10 Inertia

Physics 111: Theoretical Mechanics
Problem Set # 8

1. (10 points) Practice with Moments of Inertia. Consider a uniform, thin hollow cone (like an ice cream cone). Its pointed end is fixed but the cone is free to rotate around the pointed end; we therefore use the pointed end as the "tagged point".

(a) (3 points) Identify three orthogonal principal axes of the cone passing through this point. Explain your answer.

(b) (7 points) Using the principal axes from (a) as a coordinate basis, compute the inertia tensor components (the cone has mass M, height h, and base radius R).

a)

The cone is azimuthally symmetric, so we can pick any arbitrary rotational axis slicing straight across the flat circular base of the cone through the center of that base, and any axis that is perpendicular to it while still parallel to the surface of the base. The third axis is rotation about the line from the base of the cone to the fixed point tip.

b)

The inertia tensor is defined as having

$$|\boldsymbol{r}|^2 = (x^2 + y^2 + z^2)$$
$$\mathcal{I}_{ij} = \iiint_{vol} dm(\boldsymbol{r}^2 \delta_{ij} - \boldsymbol{r}_i \boldsymbol{r}_{j)}$$

Radially

We can now slog through finding the components.
To make the calculations easier, I define

$$x = r\cos\theta$$
$$y = r\sin\theta$$

$$\mathcal{I}_{ij} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z}(r^2\delta_{ij} - r_i r_j)\,r\,dr\,d\theta\,dz$$

First, I will start with the diagonals:

Diagonal Components

$\mathcal{I}_{xx}$:

$$\mathcal{I}_{xx} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} r^3 - r^3\cos^2\theta\,dr\,dz\,d\theta$$
$$\mathcal{I}_{xx} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} r^3(1-\cos^2\theta)dr\,dz\,d\theta$$

$$\mathcal{I}_{xx} = m\int_{\theta=0}^{2\pi}(1-\cos^2\theta)\int_{z=0}^{h}\left.\frac{r^4}{4}\right|_0^{\frac{R}{h}z} dz\,d\theta$$
$$\mathcal{I}_{xx} = m\frac{1}{4}\left(\frac{R}{h}\right)^4\int_{\theta=0}^{2\pi}(1-\cos^2\theta)\int_{z=0}^{h} z^4\,dz\,d\theta$$
$$\mathcal{I}_{xx} = m\frac{1}{4}\left(\frac{R}{h}\right)^4\int_{\theta=0}^{2\pi}(1-\cos^2\theta)\int_{z=0}^{h} z^4\,dz\,d\theta$$
$$\mathcal{I}_{xx} = m\frac{1}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\int_{\theta=0}^{2\pi}(1-\cos^2\theta)d\theta$$

$$\begin{aligned}
\mathcal{I}_{xx} &= m\frac{1}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\int_{\theta=0}^{2\pi}(1-\cos^2\theta)d\theta \\
&= m\frac{1}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\left(\theta-\frac{\sin^3\theta}{3}\right)\Big|_0^{2\pi} \\
&= m\frac{1}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}(2\pi)
\end{aligned}$$

$$\mathcal{I}_{xx} = \frac{\pi R^4 hm}{10}$$

$\mathcal{I}_{yy}$:

$$\mathcal{I}_{yy} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} r^3 - r^3\sin^2\theta\,dr\,dz\,d\theta$$

Using the result from before, this collapses to

$$\begin{aligned}
\mathcal{I}_{yy} &= \frac{m}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\int_{\theta=0}^{2\pi}(1-\sin^2\theta)d\theta \\
&= \frac{m}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\left[\left(\theta-\frac{\cos^3\theta}{3}\right)\Big|_{\theta=0}^{2\pi}\right] \\
&= \frac{m}{4}\left(\frac{R}{h}\right)^4\frac{h^5}{5}\left[2\pi-\frac{1}{3}+\frac{1}{3}\right] \\
&= \frac{m}{10}\pi R^4 h
\end{aligned}$$

$\mathcal{I}_{zz}$:

$$\begin{aligned}
\mathcal{I}_{ij} &= m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z}(r^2-z^2)\,r\,dr\,d\theta\,dz \\
&= m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} r^3 - z^2 r\,dr\,d\theta\,dz \\
&= m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\frac{r^4}{4}-\frac{z^2r^2}{2}\Big|_{r=0}^{\frac{R}{h}z} \\
&= m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\frac{\left(\frac{R}{h}\right)^4 z^4}{4}-\frac{z^4\left(\frac{R}{h}\right)^2}{2}dz\,d\theta \\
&= m\left[\frac{\left(\frac{R}{h}\right)^4}{4}-\frac{\left(\frac{R}{h}\right)^2}{2}\right]\int_{\theta=0}^{2\pi}\int_{z=0}^{h} z^4 dz\,d\theta \\
&= m\left[\frac{\left(\frac{R}{h}\right)^4}{4}-\frac{\left(\frac{R}{h}\right)^2}{2}\right]\int_{\theta=0}^{2\pi}\frac{h^5}{5}\,d\theta
\end{aligned}$$

$$\begin{aligned}
\mathcal{I}_{zz} &= \frac{2\pi m h^5}{25}\left[\frac{\left(\frac{R}{h}\right)^4}{4}-\frac{\left(\frac{R}{h}\right)^2}{2}\right] \\
&= \frac{2\pi m}{25}\left[\frac{R^4h}{4}-\frac{R^2h^4}{2}\right]
\end{aligned}$$

Off diagonals

$$\mathcal{I}_{ij} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} -r_ir_j\,r\,dr\,d\theta\,dz \qquad i\neq j$$

I_{xy}:

$$\begin{aligned}
\mathcal{I}_{xy} &= m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} -r^3\cos\theta\sin\theta\,dr\,dz\,d\theta \\
&= -m\int_{\theta=0}^{2\pi}\cos\theta\sin\theta\int_{z=0}^{h}\frac{\left(\frac{R}{h}z\right)^4}{4}dz\,d\theta \\
&= -\frac{m}{20}\int_{\theta=0}^{2\pi}\cos\theta\sin\theta R^4 h\,d\theta \\
&= -\frac{mR^4h}{20}\left(\frac{1}{2}\sin^2\theta\right)\Big|_{\theta=0}^{2\pi} \\
&= 0
\end{aligned}$$

$\mathcal{I}_{xz}$:

$$\begin{aligned}
\mathcal{I}_{xz} &= -m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} z\cos\theta\, r\,dr\,d\theta\,dz \\
&= -\frac{m}{2}\left(\frac{R}{h}\right)^2\int_{\theta=0}^{2\pi}\int_{z=0}^{h} z^2\cos\theta\,d\theta\,dz
\end{aligned}$$

We see that this integral depends on

$$\int_0^{2\pi}\cos\theta = 0$$

so $\mathcal{I}_{xz} = 0$:

$\mathcal{I}_{yz}$

$$\mathcal{I}_{ij} = m\int_{\theta=0}^{2\pi}\int_{z=0}^{h}\int_{r=0}^{\frac{R}{h}z} -z\sin\theta\, r^2\,dr\,d\theta\,dz$$

This depends on

$$\int_0^{2\pi}\sin\theta = 0$$

so $\mathcal{I}_{yz} = 0$.

The matrix must be symmetric, so $\mathcal{I}_{xy} = \mathcal{I}_{yx}, \mathcal{I}_{xz} = \mathcal{I}_{zx}, \mathcal{I}_{yz} = \mathcal{I}_{zy}$

End of problem

This gives us finally the matrix

$$\mathcal{I} = \begin{bmatrix} \frac{\pi m R^4 h}{10} & 0 & 0 \\ 0 & \frac{\pi m R^4 h}{10} & 0 \\ 0 & 0 & \frac{2\pi m}{25}\left[\frac{R^4 h}{4} - \frac{R^2 h^4}{2}\right] \end{bmatrix}$$

$$\boxed{\mathcal{I} = \pi m R^2 \begin{bmatrix} \frac{R^2 h}{10} & 0 & 0 \\ 0 & \frac{R^2 h}{10} & 0 \\ 0 & 0 & \frac{R^2 h}{50} - \frac{h^4}{25} \end{bmatrix}}$$

2. (10 points) Even More Practice with Moments of Inertia.

(a) (7 points) Consider a Toblerone bar of mass M whose two ends are equilateral triangles parallel to the xy plane with side length $2a$. The bar's centre of mass at the origin with the Toblerone bar's axis along the z axis. Find the components of the inertia tensor $\mathcal{I}_{zx}$, $\mathcal{I}_{zy}$, and $\mathcal{I}_{zz}$ (note that the first two can be found by inspection).

(b) (3 points) For the geometry in part (a), identify three orthogonal principal axes of the bar. Explain your answer.

a

Instead of considering the periodically triangular structure of Toblerone chocolate which i was doing initially (pictured below), I am considering the cartridge (thanks Conner).

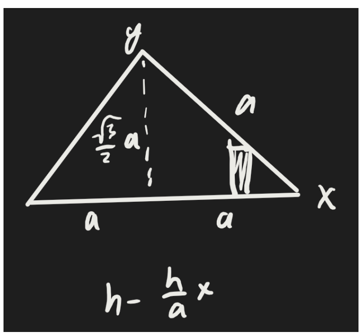
$$
\begin{aligned}
|r|^2 &= (x^2 + y^2 + z^2) \\
\mathcal{I}_{ij} &= \iiint_{vol} dm(r^2\delta_{ij} - r_i r_j)
\end{aligned}
$$

For principle axes, the inertia matrix is diagonal so $\mathcal{I}_{zx} = \mathcal{I}_{zy} = 0$.

This leaves just the $\mathcal{I}_{zz}$ term

$$
\begin{aligned}
\mathcal{I}_{zz} &= \iiint_{vol} r^2 - z^2 \, dm \\
&= M \iiint_{vol} x^2 + y^2
\end{aligned}
$$

Let the height of the bar be denoted h.
We can evaluate this.

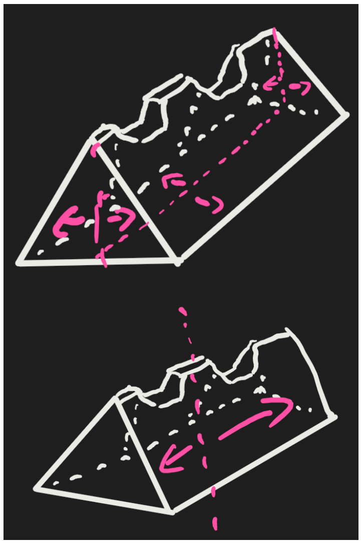


There is no z dependence, so this is just

$$
\begin{aligned}
\mathcal{I}_{zz} &= Mh \int_{y=0}^{\frac{\sqrt{3}}{2}|a-x|} \int_{x=-a}^{a} x^2 + y^2 \, dx \, dy \\
&= 2Mh \int_{x=0}^{a} \int_{y=0}^{\frac{\sqrt{3}}{2}a - x} x^2 + y^2 \, dy \, dx \\
&= 2Mh \int_{x=0}^{a} x^2 \left(\frac{\sqrt{3}}{2}(a-x) \right) + \frac{\left(\frac{\sqrt{3}}{2}(a-x)\right)^3}{3} \, dx \\
&= \frac{7Mha^4}{16\sqrt{3}}
\end{aligned}
$$

b

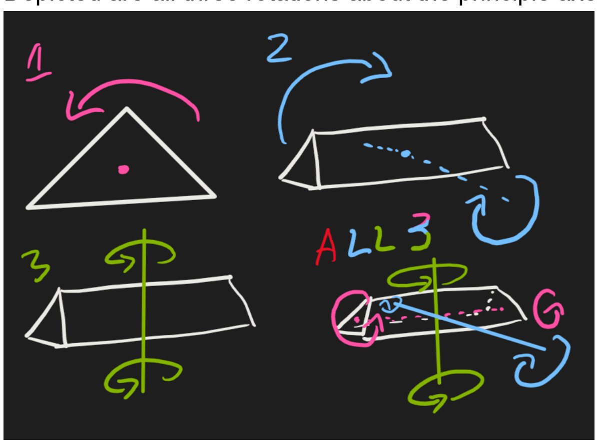
There are two reflection symmetries in the system, corresponding to the equilateral if we let the Toblerone have pyramids. These are triangle/rectangular base and flip across the length, since the uneven top is still symmetric on either side of this plane.



If we assume that the Toblerone has not been eaten and is in a nice triangular prism and has no ridges, it is also rotationally symmetric

These give two of the three axis. The third is rotation about the center line of the bar.

Depicted are all three rotations about the principle axes:



3. (16 points) Set of Masses. Three identical point masses m are arranged at the following positions in a Cartesian coordinate system: $a\,\hat{x}$, $a\,\hat{y} + 2a\,\hat{z}$, and $2a\,\hat{y} + a\,\hat{z}$ for some constant a.

(a) (5 points) Find the inertia tensor, $\mathcal{I}$, in the given Cartesian coordinate system.

(b) (7 points) Find the principal moments and a set of orthogonal principal axes.

(c) (4 points) We start the masses spinning with angular velocity $\vec{\omega} = \omega_0\,\hat{z}$, assuming they are held rigidly together in the same shape. Find the angular momentum, $\vec{\ell}$. If no external force is applied to the system, which of the following do we expect to remain constant as a function of time: $\vec{\ell}$, $\vec{\omega}$, both, or neither? Explain your answer

a

The coordinates are of the three masses are given by the following:

$$\mathbf{R} = a\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 2\\ 0 & 2 & 1\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}$$

$$\text{with}$$

$$\mathbf{M} = \begin{pmatrix}m\\ m\\ m\end{pmatrix}$$

$$|r|^2 = (x^2 + y^2 + z^2)$$

$$\mathcal{I}_{ij} = \iiint_{vol} dm(r^2\delta_{ij} - r_i r_j)$$

We can find the easy diagonals by inspection:

$$\begin{aligned}\mathcal{I}_{xy} &= \iiint_{vol} -xy\,dm \\ &= -\sum_{i=1}^{3} \mathbf{R}_{x,i}\mathbf{R}_{y,i}m_i \\ &= 0\end{aligned}$$

Similarly,

$$\mathcal{I}_{xz} = -\sum_{i=1}^{3} R_{x,i}R_{z,i}m_i = 0$$

$$\mathcal{I}_{yz} = 0 + 2a^2 + 2a^2 = -4ma^2$$

Now we can find the diagonals

$$\begin{aligned}\mathcal{I}_{yy} &= m\iiint_{vol} x^2 + z^2 \\ &= m(a^2 + 4a^2 + a^2) \\ &= 6ma^2\end{aligned}$$

$$\begin{aligned}\mathcal{I}_{zz} &= m\iiint_{vol} x^2 + y^2 \\ &= 6ma^2\end{aligned}$$

$$\begin{aligned}\mathcal{I}_{xx} &= \iiint_{vol} y^2 + z^2 \\ \mathcal{I}_{xx} &= m(a^2 + 4a^2 + 4a^2 + a^2) \\ &= 10ma^2\end{aligned}$$

Final Matrix

$$\mathcal{I} = \begin{bmatrix} 10ma^2 & 0 & 0 \\ 0 & 6ma^2 & -4ma^2 \\ 0 & -4ma^2 & 6ma^2 \end{bmatrix}$$
$$\mathcal{I} = 2ma^2 \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & -2 & 3 \end{bmatrix}$$

b

(b) (7 points) Find the principal moments and a set of orthogonal principal axes.

We found that

$$\mathcal{I} = 2ma^2 \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & -2 & 3 \end{bmatrix}$$

The principle axes are the eigenvectors of this matrix, with the principle moments as the eigenvalues.

We can solve this easily.

$$\det\left(\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & -2 & 3 \end{bmatrix} - \lambda I\right) = 0$$
$$\det\begin{bmatrix} 5-\lambda & 0 & 0 \\ 0 & 3-\lambda & -2 \\ 0 & -2 & 3-\lambda \end{bmatrix} = 0$$

This leads to
$(5-\lambda)[(3-\lambda)(3-\lambda)-4]=0$
We see that $\lambda=5$ is the first eigenvalue.

$$(3-\lambda)^2-4=0$$
$$5-6\lambda+\lambda^2=0$$

This is factorable into

$$(\lambda-1)(\lambda-5)=0$$

And has two eigenvalue solutions - one is degenerate.

$$\lambda_1=1,\, \lambda_2=5, \lambda_3=5$$

Eigenvectors must satisfy

$$\begin{pmatrix} 5-\lambda & 0 & 0 \\ 0 & 3-\lambda & -2 \\ 0 & -2 & 3-\lambda \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\vec{0}$$

For λ_1, this is

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{bmatrix}$$

We get the expression

$$x=0$$
$$y=z$$

so

$$E_1=\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

For λ_2, λ_3 this is

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & -2 \\ 0 & -2 & -2 \end{bmatrix}$$

Which yields
$y=-z.$
x can be anything. We must find two linearly independent eigenvectors
so

$$E_2=\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

A nice vector that is orthogonal to this is

$$E_3=\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

This gives the basis for the principle axes of rotation as

$$\mathbb{B} = 2ma^2 \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix}$$

with principle moments of inertia
$2ma^2$ and $10ma^2$.

c

(c) (4 points) We start the masses spinning with angular velocity $\vec{\omega} = \omega_0 \hat{z}$, assuming they are held rigidly together in the same shape. Find the angular momentum, $\vec{\ell}$. If no external force is applied to the system, which of the following do we expect to remain constant as a function of time: $\vec{\ell}$, $\vec{\omega}$, both, or neither? Explain your answer

The angular momentum is given by

$$\vec{L} = \mathcal{I}\mathrm{w}$$
$$= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ \omega_0 \end{pmatrix}$$
$$\vec{L} = 2ma^2\omega_0 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

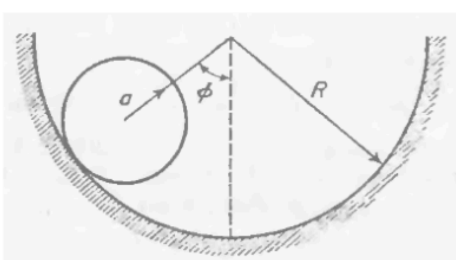
With no external forces, the angular momentum will be conserved. Because this is in the principle axis coordinate system and each vector is orthogonal, there are no cross interaction terms for the momentum to transfer between, so both $\vec{l}$ and $\vec{\omega}$ will remain constant.

4. (14 points) Rolling Oscillator. A uniform circular disk of radius a and mass M is constrained to roll without slipping on a semicircular surface of radius R as indicated in the figure. The motion is restricted to the plane indicated in the figure, and the angle of the centre of the disk relative to the vertical is denoted by ϕ.

(a) (3 points) Argue that the vector perpendicular to the plane is a principal axis, and calculate the corresponding principal moment about the disk's CM.

(b) (7 points) Determine the kinetic energy of the disk. *Hint: consider both translational and rotational motion, and be sure to use the fact that the disk is rolling without slipping.*

(c) (4 points) Assuming that gravity points down towards the bottom of the surface, determine the angular frequency of small oscillations about the bottom of the bowl.



a

The ball rolls in a circular motion about the center of the circle, always at the same distance. This fixed point at the center is a useful reference frame because it is stuck and the motion is at a fixed radius away - rigid reference points with rotational symmetry are nice. The perpendicular axis to the plane placed at the center of the path circle therefore makes sense as a principle axis.

b

When rolling without slipping, the ball has an instantaneous pivot point at the point of contact. However, to easily consider energy, I will translate this the be about the axis at the center with the parallel axis theorem.

The moment of inertia about this disks center of mass is $\frac{1}{2}MR^2$.

$$\begin{aligned} T &= \frac{1}{2}I_{cm}\dot{\phi}^2 + \frac{1}{2}Mv_{cm}^2 \\ &= \frac{1}{4}MR^2\dot{\phi}^2 + \frac{1}{2}M\Big((R-\alpha)\dot{\phi}\Big)^2 \\ &= \frac{1}{4}MR^2\dot{\phi}^2 + \frac{1}{2}M(R-\alpha)^2\dot{\phi}^2 \\ T &= \dot{\phi}^2\left(\frac{MR^2}{4} + \frac{MR^2}{2} + \frac{M\alpha^2}{2} - MR\alpha\right) \\ T &= \dot{\phi}^2 M\left(\frac{3}{4}R^2 + \frac{\alpha^2}{2} - \alpha R\right) \end{aligned}$$

C

The ball has gravitational potential about the center of mass, which is given by

$$\begin{aligned} U &= mgh \\ &= mg(R - (R-\alpha)\cos\phi) \\ &= mg(R - R\sin\phi + \alpha\cos\phi) \\ &= mgR(1-\cos\phi) + mg\alpha\cos\phi \end{aligned}$$

This gives the Lagrangian as

$$\mathscr{L} = \dot{\phi}^2 M\left(\frac{3}{4}R^2 + \frac{\alpha^2}{2} - \alpha R\right) - mgR(1-\cos\phi) - mg\alpha\cos\phi$$

We can grab the EL equations from this to find the frequency of oscillation.

$$\frac{\partial\mathscr{L}}{\partial\phi} = \frac{d}{dt}\frac{\partial\mathscr{L}}{\partial\dot{\phi}}$$

$$\begin{aligned} \frac{\partial\mathscr{L}}{\partial\phi} &= -Mg(R-\alpha)\sin\phi \\ \frac{d}{dt}\frac{\partial\mathscr{L}}{\partial\dot{\phi}} &= 2M\left(\frac{3}{4}R^2 + \frac{\alpha^2}{2} - \alpha R\right)\ddot{\phi} \end{aligned}$$

So we have

$$\ddot{\phi} = \frac{-g(R-\alpha)}{\frac{3}{2}R^2 + \alpha^2 - 2\alpha R}\sin\phi$$

For small angles, we can approximate $\sin\phi = \phi$
This gives us an equation like

$$\ddot{\phi} = -\omega^2\phi$$

so

$$\omega = \sqrt{\frac{g(R-\alpha)}{\frac{3}{2}R^2 + \alpha^2 - 2\alpha R}}$$